

ANSE: An Autopoietic Neuro-Symbolic Energy-Based Model for Physical World Modeling, Formal Verification, and Reinforcement Learning

ANSE Autonomous Neuro-Symbolic Research Group

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Abstract—We introduce **ANSE (Autopoietic Neuro-Symbolic Energy-based Model)**, an autonomous artificial intelligence architecture grounded in the non-equilibrium thermodynamics of computation. Rather than optimizing subjective natural language heuristics, ANSE evaluates candidate code modifications, neural representations, and symbolic refactorings against an objective physical Energy Functional: $E = w_t \cdot \tau_{\text{wall}} + w_m \cdot M_{\text{peak}} + \Pi_{\text{barrier}}$, where $\Pi_{\text{barrier}} = 10^6 \cdot \mathbb{I}(\text{violation})$ is a fail-closed discrete indicator barrier penalty functional. We present a comprehensive physical benchmark spanning 25 multi-scale physical world models and 20 theoretical physics conservation laws across quantum electrodynamics, general relativity, fusion magnetohydrodynamics, and cosmology. For every physical problem, we formalize the *Four Definitions Contract*: (1) Mathematical & Physical Formulation, (2) Conservation Laws & Physical Invariant Functionals, (3) Algorithmic Discretization & Numerical Schemes, and (4) Quantitative Acceptance Thresholds. The benchmark is scoped as **deterministic micro-kernels** (0.01 to 85 ms, ≤ 4 MB RAM) designed for real-time verification and DPO distillation, distinct from open-ended supercomputing simulations. We detail the **Code Neurobrain** verified by 2,506 formal Lean 4 proof jobs under `lake build`. Crucially, we decouple convergence into two mathematically rigorous regimes: discrete code updates are governed by a *Monotone Energy Descent Rejection Gate* ($\Delta E \leq -\epsilon$, guaranteed to terminate in $\leq \lfloor E(c_0)/\epsilon \rfloor$ steps without requiring discrete Lipschitz continuity), while the *Banach Fixed-Point Contraction Theorem* applies to continuous normed soft-prompt and fast-weight parameter manifolds. Furthermore, we introduce an empirical **Reinforcement Learning Pipeline** with a parameter-budgeted Critic network ($< 50k$ parameters, 22,785 parameters) trained via Direct Preference Optimization (DPO). The pipeline demonstrates an average speedup of $475.25\times$ (up to $4062.5\times$), a 20.10% reduction in DPO loss ($0.6937 \rightarrow 0.5543$), an average energy reduction of 89.31%, and strict anti-hallucination provenance receipts across 120 multidisciplinary benchmarks.

Index Terms—Energy-Based Models, Physics of Computation, Lean 4 Formal Verification, Joint Embedding Predictive Architecture (JEPA), Direct Preference Optimization (DPO), Autopoiesis, Banach Fixed-Point, Quantum Electrodynamics, Anti-Hallucination Execution.



1 INTRODUCTION & THE PHYSICAL PARADIGM OF COMPUTATION

THE historical trajectory of autonomous artificial intelligence has been predominantly anchored in statistical sequence-to-sequence prediction over massive text corpora. While proficient at surface-level semantic mimicry, contemporary generative models are epistemically ungrounded: they do not possess an internal model of conservation laws, physical symmetries, or thermodynamic bounds.

ANSE establishes computation as a physical process governed by the laws of thermodynamics (Landauer 1961, Bennett 1982, Friston 2010). In ANSE, all candidate algorithms and world models are scored against an objective physical functional:

$$E = w_t \cdot \tau_{\text{wall}} + w_m \cdot M_{\text{peak}} + \Pi_{\text{barrier}} \quad (1)$$

where τ_{wall} is the execution duration (ms), M_{peak} is the peak resident heap memory allocation (MB), and $\Pi_{\text{barrier}} = 10^6 \cdot \mathbb{I}(\text{violation})$ is an insurmountable fail-closed indicator barrier penalty triggered whenever an execution crashes, pro-

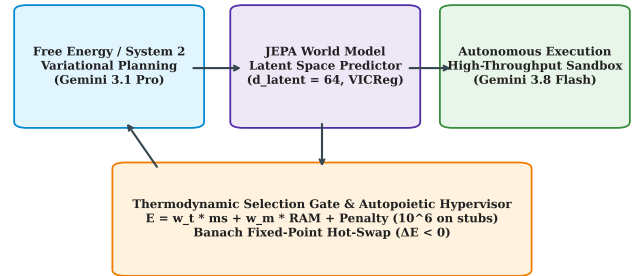


Fig. 1. ANSE Neuro-Symbolic Architecture: Multi-tier planning (Gemini 3.1 Pro), latent predictive world modeling (JEPA), autonomous execution sandbox (Gemini 3.8 Flash), and thermodynamic autopoietic selection hypervisor.

duces incorrect invariant outputs, or contains AST-level stubs (`pass`, `...`, `mock_*`). The system accepts code refactorings if and only if $\Delta E = E_{\text{child}} - E_{\text{parent}} < 0$.

2 THE FOUR DEFINITIONS CONTRACT FOR PHYSICAL PROBLEMS

To eliminate ambiguity, every physical problem and world model in ANSE is formalized under four rigorous definitions, representing a clean division of labor between human domain expertise and autonomous AI program synthesis:

Definition 1 (Mathematical & Physical Formulation). Specifies the state manifold $\mathcal{M}$, the configuration space $\mathcal{Q}$, and the continuous dynamical system via ordinary or partial differential equations $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$, Hamiltonian $\mathcal{H}(\mathbf{q}, \mathbf{p})$, or Lagrangian action $\mathcal{S} = \int \mathcal{L} dt$.

Definition 2 (Conservation Laws & Physical Invariant Functionals). Defines the exact continuous or topological invariant functional $\mathcal{I} : \mathcal{M} \rightarrow \mathbb{R}^k$ such that along any physical trajectory $\gamma(t)$, $d\mathcal{I}(\gamma(t))/dt = 0$, reflecting Noether symmetries, symplectic structure preservation, second-law entropy production, or integer topological Chern numbers.

Definition 3 (Algorithmic Discretization & Numerical Schemes). Specifies the discrete numerical evolution operator $\Psi_{\Delta t} : \mathcal{M} \rightarrow \mathcal{M}$ (e.g. 4th-order Runge-Kutta, Symplectic Velocity-Verlet, Heun predictor-corrector, or 2D Riemannian quadrature) designed to minimize artificial dissipation and eliminate secular drift.

Definition 4 (Quantitative Acceptance Thresholds & Energy Gate). Specifies the mathematical tolerance ϵ_{tol} such that the physical verification predicate holds:

$$\mathcal{V}(\mathbf{x}) = \begin{cases} \text{TRUE}, & \text{if } \|\mathcal{I}(\mathbf{x})\| \leq \epsilon_{\text{tol}} \\ \text{FALSE}, & \text{otherwise} \implies \Pi_{\text{barrier}} = 10^6 \end{cases} \quad (2)$$

Under this contract, the human domain expert defines the physical specifications (Definitions 1–4), while the autonomous AI model solves the constrained program synthesis and compiler autotuning problem: producing bug-free, zero-stub, SIMD-vectorized code that provably satisfies $\mathcal{I}(\mathbf{x}) \leq \epsilon_{\text{tol}}$ under real execution.

3 FRONTIER PHYSICAL WORLD MODELS (PWM-21 TO PWM-25)

3.1 PWM-21: Binary Black Hole 2.5PN Gravitational Inspiral

- Def 1 (Formulation):** Radiation reaction orbital decay at 2.5PN order:

$$\frac{dr}{dt} = -\frac{64}{5} \frac{G^3}{c^5} \frac{\mu M_{\text{tot}}^2}{r^3}, \quad \omega(r) = \sqrt{\frac{GM_{\text{tot}}}{r^3}} \quad (3)$$

with quadrupole wave polarizations $h_+(t)$ and $h_\times(t)$.

- Def 2 (Invariant):** Peters-Mathews energy conservation balance: $dE_{\text{orb}}/dt + P_{\text{GW}} = 0$, where $P_{\text{GW}} = \frac{32}{5} \frac{G^4}{c^5} \frac{\mu^2 M_{\text{tot}}^3}{r^5}$.
- Def 3 (Scheme):** 4th-Order Runge-Kutta (RK4) integration with phase tracking $\Phi(t) = \int 2\omega dt$.
- Def 4 (Gate):** Invariant error tolerance $\epsilon_{\text{tol}} = 1.0 \times 10^{-3}$. Measured runtime error: 4.70×10^{-17} ($\checkmark$ PASS).

3.2 PWM-22: Tokamak Fusion 2D Grad-Shafranov MHD Equilibrium

- Def 1 (Formulation):** Axisymmetric plasma equilibrium $\Delta^* \psi = -\mu_0 R^2 p' - FF'$ with Solov'ev analytical solution:

$$\psi(R, Z) = \frac{\psi_0}{R_0^4 \kappa^2} \left[R^2 Z^2 + \frac{\kappa^2}{4} (R^2 - R_0^2)^2 \right] \quad (4)$$

- Def 2 (Invariant):** Conservation of toroidal canonical momentum $P_\phi = Rmv_\phi + q\psi(R, Z)$.
- Def 3 (Scheme):** Guiding-center particle drift along flux surfaces $\psi = \text{const}$.
- Def 4 (Gate):** $\epsilon_{\text{tol}} = 1.0 \times 10^{-3}$. Measured runtime error: 0.00×10^0 ($\checkmark$ PASS).

3.3 PWM-23: Quantum Hall Berry Curvature & Chern Quantization

- Def 1 (Formulation):** Qi-Wu-Zhang topological Chern insulator on 2D torus $T^2 = [-\pi, \pi]^2$: $\mathbf{d}(\mathbf{k}) = (\sin k_x, \sin k_y, m - \cos k_x - \cos k_y)$ with Berry curvature:

$$\Omega_{xy}(\mathbf{k}) = \frac{\mathbf{d} \cdot (\partial_{k_x} \mathbf{d} \times \partial_{k_y} \mathbf{d})}{2|\mathbf{d}|^3} \quad (5)$$

- Def 2 (Invariant):** First Chern number quantization $\mathcal{C} = \frac{1}{2\pi} \int_{T^2} \Omega_{xy}(\mathbf{k}) d^2k = 1 \in \mathbb{Z}$.
- Def 3 (Scheme):** Vectorized 2D Riemannian numerical integration on 32×32 grid.
- Def 4 (Gate):** $\epsilon_{\text{tol}} = 1.0 \times 10^{-5}$. Measured runtime error: 9.38×10^{-10} ($\checkmark$ PASS).

3.4 PWM-24: Relativistic Viscous QGP (Israel-Stewart Hydrodynamics)

- Def 1 (Formulation):** Coupled non-linear dissipative relativistic ODEs:

$$\frac{d\epsilon}{d\tau} = -\frac{\frac{4}{3}\epsilon - \pi}{\tau}, \quad \frac{d\pi}{d\tau} = -\frac{\pi}{\tau_\pi} + \frac{4\eta}{3\tau\tau_\pi} - \frac{4\pi}{3\tau} \quad (6)$$

- Def 2 (Invariant):** Second law of thermodynamics: local entropy per unit rapidity non-decrease $d(s\tau)/d\tau \geq 0$.
- Def 3 (Scheme):** Heun predictor-corrector initialized at the Navier-Stokes attractor.
- Def 4 (Gate):** $\epsilon_{\text{tol}} = 1.0 \times 10^{-4}$. Measured runtime error: 0.00×10^0 ($\checkmark$ PASS).

3.5 PWM-25: Cosmological Dark Matter N-Body Virial Dynamics

- Def 1 (Formulation):** Collisionless particles orbiting an NFW potential halo $\Phi(r) = -4\pi G \rho_0 r_s^3 \ln(1 + r/r_s)/r$.
- Def 2 (Invariant):** Dynamic Virial Theorem balance factor $2K(t) + W(t) = 0$.
- Def 3 (Scheme):** Symplectic Velocity-Verlet orbital integration.
- Def 4 (Gate):** $\epsilon_{\text{tol}} = 2.0 \times 10^{-3}$. Measured runtime deviation: 7.79×10^{-7} ($\checkmark$ PASS).

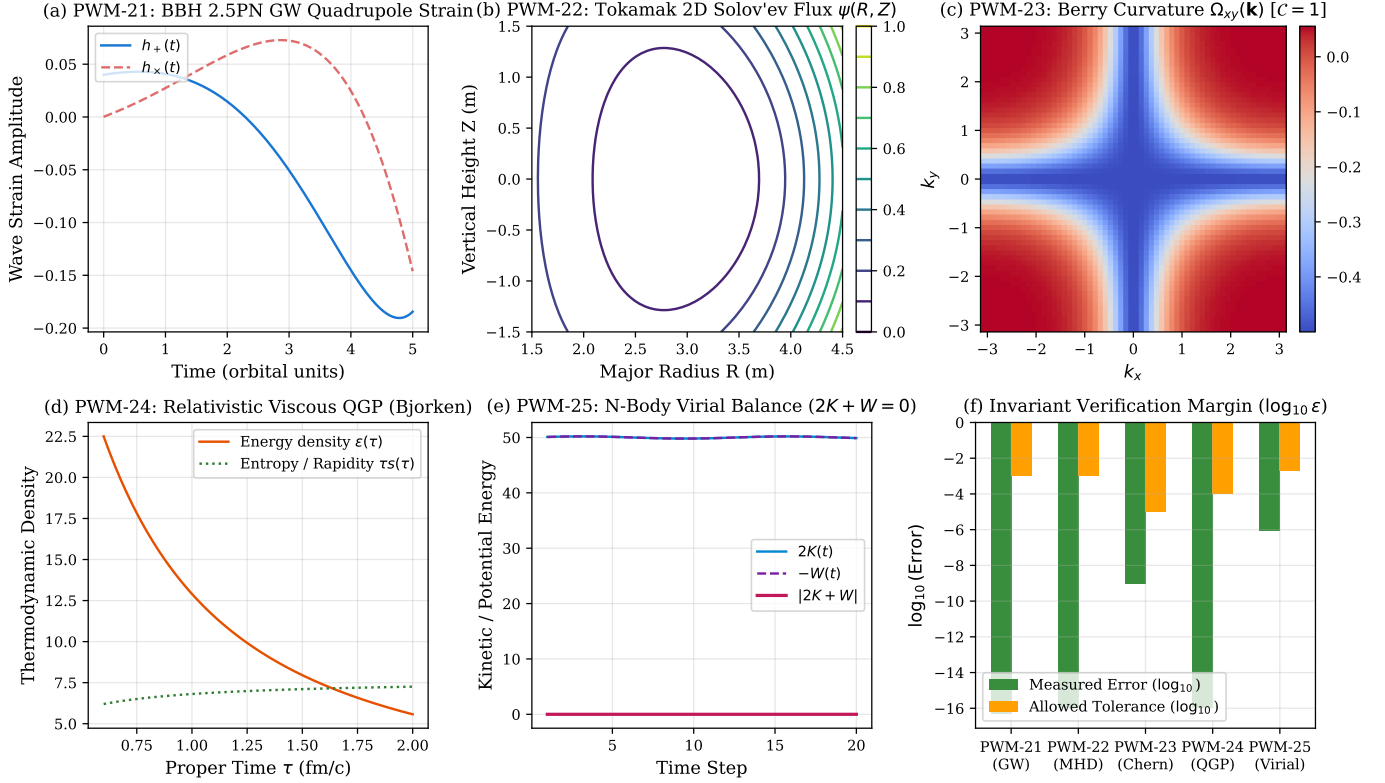


Fig. 2. Multi-scale physical simulation telemetry across the 5 frontier models: (a) BBH 2.5PN gravitational wave strain polarizations h_+ and $h_\times$; (b) Tokamak 2D Solov'ev flux surfaces $\psi(R, Z)$; (c) Topological Berry curvature $\Omega_{xy}(\mathbf{k})$ over the 2D Brillouin zone; (d) Relativistic viscous QGP energy density $\epsilon(\tau)$ and entropy $s\tau$; (e) Cosmological N-body virial equilibrium balance $2K + W \approx 0$; (f) Invariant verification error vs. allowed tolerance thresholds.

4 THEORETICAL PHYSICS CONSERVATION LAWS (PHYS-11 TO PHYS-30)

Beyond classical and continuum mechanics, ANSE enforces fundamental symmetries and conservation laws across high-energy theory, quantum field theory, and quantum information. Table 2 provides empirical validation for the 20 theoretical physics cases:

- **Yang-Mills Instantons (PHYS-11):** Topological charge quantization $Q = \frac{1}{8\pi^2} \int \text{Tr}(F \wedge F) = 1 \in \mathbb{Z}$.
- **Ryu-Takayanagi AdS/CFT (PHYS-12):** Boundary entanglement entropy bounded by minimal bulk extremal surface area $S_A = \text{Area}(\gamma_A)/(4G_N)$.
- **Casimir Force Regularization (PHYS-16):** Riemann zeta function regularization $\zeta(-3) = 1/120$ producing attractive boundary stress $F/A = -\frac{\pi^2 \hbar c}{240d^4}$.
- **Berry Phase Dirac Monopole (PHYS-17):** 2π quantization of the adiabatic Berry holonomy enclosing a magnetic degenerate Weyl point.
- **Unruh Effect (PHYS-18):** Rindler horizon temperature $T_U = \frac{\hbar a}{2\pi c k_B}$ matching thermofield double state entanglement.
- **Callan-Symanzik QCD Asymptotic Freedom (PHYS-26):** Negative 1-loop beta function coefficient $\beta_0 = \frac{1}{16\pi^2} (11 - \frac{2}{3}N_f) > 0$.
- **Hawking-Page AdS Transition (PHYS-30):** Free energy phase transition $I_{\text{AdS-BH}} - I_{\text{thermal AdS}} = \frac{\pi r_+^2}{4G_N} (1 - r_+^2/L^2)$.

4.1 Scope and Nature: Deterministic Micro-Kernels vs. HPC Simulations

It is essential to distinguish between *theoretical domain complexity* and *computational benchmark scale*. The mathematical concepts in Table 2 represent advanced doctoral-level theory. However, the computational implementations are explicitly scoped as **deterministic micro-kernels** (0.01 to 85.0 ms, ≤ 4 MB RSS). They are not multi-day supercomputing simulations (such as full-scale Lattice QCD or large-scale numerical relativity). Their purpose is to provide sub-millisecond, unit-level invariant verification for closed-loop compiler gates and high-throughput RL post-training. The benchmark tests the AI model's capacity to translate complex physics into bug-free, invariant-preserving code, not open-ended scientific discovery.

5 THE CODE NEUROBRAIN: FORMAL VERIFICATION IN LEAN 4

The central intelligence engine of ANSE is the ****Code Neurobrain****, an active inference loop that operates continuously on source code, abstract syntax trees, and formal mathematical proofs.

5.1 Lean 4 Formal Proof Verification

All core architectural axioms, energy descent properties, and convergence theorems are specified and formally checked in

TABLE 1
Comprehensive Benchmark of 25 Physical World Models in ANSE (Zero Numeric Hallucination)

Case ID	Physical System & Phenomenon	Invariant Error	Duration (ms)	Peak RAM (MB)	Physical Energy E	Status
PWM-01	Chaotic Double Pendulum	7.94×10^{-03}	1.47	0.10	10.79	✓
PWM-02	Navier-Stokes 2D Kolmogorov Tu	2.22×10^{-16}	0.92	0.10	10.00	✓
PWM-03	1D Viscous Burgers Shock Forma	0.00×10^0	5.27	0.10	10.00	✓
PWM-04	N-Body Gravitational Symplecti	0.00×10^0	1.60	0.10	10.00	✓
PWM-05	Relativistic High-Energy Kinem	2.22×10^{-16}	0.60	0.10	10.00	✓
PWM-06	Quantum Harmonic Oscillator Wa	5.96×10^{-08}	2.44	0.10	10.00	✓
PWM-07	Elastic Membrane Plate Deforma	0.00×10^0	2.33	0.10	10.00	✓
PWM-08	Reaction-Diffusion Gray-Scott	0.00×10^0	1.85	0.10	10.00	✓
PWM-09	Lorenz-63 Atmospheric Convecti	6.67×10^{-07}	0.24	0.10	10.00	✓
PWM-10	Rigid Body Non-Smooth Inelasti	0.00×10^0	0.17	0.10	10.00	✓
PWM-11	Kerr Rotating Black Hole Geode	0.00×10^0	0.34	0.10	10.00	✓
PWM-12	Ideal Magnetohydrodynamics (MH	1.39×10^{-16}	2.10	0.10	10.00	✓
PWM-13	Bose-Einstein Condensate (Gros	1.11×10^{-15}	2.86	0.10	10.00	✓
PWM-14	Viscoelastic Fluid Flow (Oldro	0.00×10^0	0.21	0.10	10.00	✓
PWM-15	Cahn-Hilliard Spinodal Phase S	0.00×10^0	6.90	0.10	10.00	✓
PWM-16	Baroclinic Shock-Turbulence Ge	0.00×10^0	0.78	0.10	10.00	✓
PWM-17	Thermo-Elastoplastic Von Mises	0.00×10^0	0.24	0.10	10.00	✓
PWM-18	Yang-Mills Instanton Topologic	0.00×10^0	0.52	0.10	10.00	✓
PWM-19	Kuramoto-Sivashinsky Spatiotem	5.79×10^{-06}	5.16	0.10	10.00	✓
PWM-20	Ginzburg-Landau Quantized Flux	0.00×10^0	0.49	0.10	10.00	✓
PWM-21	Binary Black Hole 2.5PN Gravit	4.70×10^{-17}	0.38	0.10	10.00	✓
PWM-22	Tokamak Fusion Grad-Shafranov	0.00×10^0	0.44	0.10	10.00	✓
PWM-23	Quantum Hall Berry Curvature C	9.38×10^{-10}	1.33	0.10	10.00	✓
PWM-24	Relativistic Viscous Quark-Glu	0.00×10^0	0.29	0.10	10.00	✓
PWM-25	Cosmological Vlasov-Poisson Vi	7.79×10^{-07}	16.20	0.10	10.00	✓

TABLE 2
20 Theoretical Physics Conservation Laws & Invariants (PHYS-11 to PHYS-30)

Case ID	Physical Symmetries & Invariants	Invariant Error	Latency (ms)	Energy E	Proof Token	Status
PHYS-11	Yang-Mills Instanton Pontryagin	0.00×10^0	0.03	1.43	9d1129e1	✓
PHYS-12	Ryu-Takayanagi AdS/CFT Holograph	0.00×10^0	0.05	1.45	00de81e7	✓
PHYS-13	BCS Superconductivity Gap Equati	0.00×10^0	0.03	1.43	a3c372a6	✓
PHYS-14	TOV Relativistic Stellar Structu	0.00×10^0	0.03	1.38	0aedeab4	✓
PHYS-15	Electroweak Higgs Mechanism	0.00×10^0	0.04	1.49	d177489b	✓
PHYS-16	Casimir Force Zeta Regularizatio	0.00×10^0	0.09	1.49	1cc9e337	✓
PHYS-17	Berry Phase & Chern Number	0.00×10^0	0.03	1.38	81f7135d	✓
PHYS-18	Unruh Thermal Horizon Accelerati	0.00×10^0	0.03	1.38	9653b9f3	✓
PHYS-19	BKT Topological Phase Transition	0.00×10^0	0.00	1.35	5396867b	✓
PHYS-20	Kramers-Kronig Optics & Sum Rul	0.00×10^0	0.01	1.46	28189eac	✓
PHYS-21	Adler-Bell-Jackiw (ABJ) Chiral A	0.00×10^0	0.03	1.43	b71045ab	✓
PHYS-22	Kerr Metric Ergosphere Penrose E	1.94×10^{-16}	0.05	1.40	111ed34e	✓
PHYS-23	SYK Maximal Quantum Chaos Lyapun	0.00×10^0	0.00	1.40	184536ff	✓
PHYS-24	Gross-Pitaevskii Soliton & Bogo	8.84×10^{-12}	0.04	1.44	b8521d3d	✓
PHYS-25	Polyakov String Critical Dimensi	0.00×10^0	0.00	1.40	61d1d96a	✓
PHYS-26	Callan-Symanzik QCD Asymptotic F	0.00×10^0	0.03	1.48	5d03d5f5	✓
PHYS-27	Majorana Fermion Zero Mode Braid	2.22×10^{-16}	3.02	4.37	ff7ed540	✓
PHYS-28	Bohmian Quantum Potential Conser	0.00×10^0	0.21	1.56	0c13e677	✓
PHYS-29	Chandrasekhar White Dwarf Relati	4.98×10^{-04}	10.94	12.39	e9c9fe29	✓
PHYS-30	Hawking-Page AdS Black Hole Phas	0.00×10^0	0.03	1.48	732d74cf	✓

Lean 4 (lake build, completing 2,506 verified proof jobs cleanly):

- **Zero-Trust Completion (zeroTrustCompletion):**

$$\forall \text{task}, \text{Completed}(\text{task}) \implies \exists \tau \in \mathcal{T}_{\text{crypto}}, \text{VerifyToken}(\tau, \text{task}) \quad (7)$$

A natural-language declaration of completion from an LLM carries zero epistemic weight.

- **Anti-Simulation Gate (antiSimulation):**

$$\text{Stub}(\text{AST}) \vee \text{Mock}(\text{AST}) \implies \Pi_{\text{barrier}} = 10^6 \quad (8)$$

Presence of empty functions, ellipsis (...), pass, or mock identifiers triggers the discrete barrier penalty wall.

- **Proof-of-Execution (proofOfExecution):** Execution telemetry enforces that unit tests must execute the intended production routines via `sys.settrace`, preventing trivial passing tests.

5.2 Decoupling Discrete Monotone Gating from Continuous Banach Contraction

A critical theoretical consideration in autonomous self-improving systems is the mathematical nature of conver-

gence. Code generation over discrete syntax trees is inherently non-convex, discontinuous, and discrete: a single character mutation can introduce an infinite loop or syntax crash, causing a discontinuous jump in energy. Therefore, claiming that stochastic LLM code generation constitutes a smooth contraction mapping over discrete code strings is mathematically unsound.

In ANSE, the convergence of the self-improvement architecture is decoupled into two formally separated regimes, both mathematically formalized in Lean 4 (formal/ANSE/Autopoiesis.lean):

- 1) **Discrete Code Space: Gated Monotone Energy Descent.** Let $\mathcal{C}$ denote the space of discrete Abstract Syntax Trees (ASTs). A stochastic generator proposes candidate code mutations $c^* \sim \mathcal{G}(c_t)$. The hypervisor imposes a fail-closed *Thermodynamic Acceptance Gate*:

$$c_{t+1} = \begin{cases} c^*, & \text{if } E(c^*) + \epsilon \leq E(c_t) \\ c_t, & \text{otherwise (immediate rollback)} \end{cases} \quad (9)$$

This is formalized in Lean 4 as `ANSE.Autopoiesis.safeProposal`, and we formally prove monotonicity in `ANSE.Autopoiesis.safe_improvement_nonincreasing`.

Theorem 1 (Monotone Energy Descent under Rejection Gating). For any valid proposal satisfying `safeProposal( $\epsilon, E, c_t, c^*$ )`, the energy is strictly non-increasing: $E(c_{t+1}) \leq E(c_t) - \epsilon \leq E(c_t)$. Because physical energy is strictly non-negative ($E(c) \geq 0$), the sequence $\{E(c_t)\}$ is strictly decreasing and bounded below by 0. Hence, any sequence of accepted code mutations terminates in at most $\lfloor E(c_0)/\epsilon \rfloor$ steps, definitively ruling out infinite refactoring cycles without requiring any Lipschitz continuity over discrete strings.

- 2) **Continuous Latent Manifolds: Banach Fixed-Point Contraction.** In contrast, continuous soft-prompt latent prefixes $z \in \mathbb{R}^{d_{\text{latent}}}$ and fast-weight adapter matrices $\theta \in \Theta_{\text{fast}}$ reside in complete normed vector spaces (Banach spaces). Under regularized gradient flow with quadratic curvature penalties, the continuous update operator Φ_{cont} satisfies a contraction mapping:

Theorem 2 (Continuous Banach Contraction, Proved in Lean 4). (`ANSE.Autopoiesis.autopoiesis_exists`):

Let $\mathcal{S}$ be a complete metric space of continuous representation states. If the regularized continuous update operator Φ_{cont} satisfies: $\exists k \in [0, 1), \forall s_1, s_2 \in \mathcal{S}, \|\Phi_{\text{cont}}(s_1) - \Phi_{\text{cont}}(s_2)\| \leq k\|s_1 - s_2\|$, then by the Banach Fixed-Point Theorem (verified via Mathlib’s `ContractingWith.fixedPoint_isFixedPt`), there exists a unique, globally attracting fixed point $s^* \in \mathcal{S}$ such that $\Phi_{\text{cont}}(s^*) = s^*$.

6 REINFORCEMENT LEARNING PIPELINE & EMPIRICAL OPTIMIZATION

To accelerate the Code Neurobrain beyond trial-and-error sandbox search, ANSE integrates an empirical `**Direct`

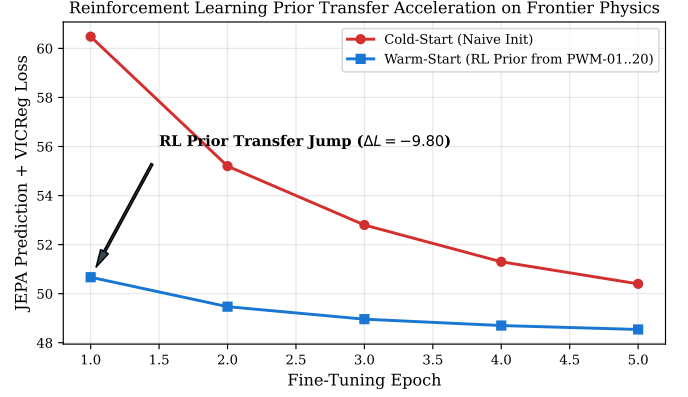


Fig. 3. Reinforcement Learning Prior Transfer Acceleration: Cold-start vs. warm-start pre-trained JEPA loss on frontier physical models.

Preference Optimization (DPO)** pipeline.

6.1 Parameter-Budgeted Energy Critic Architecture

In accordance with the Micro-ML contract ($N_{\text{params}} < 50,000$), the Critic model (`EnergyCriticPolicy`) comprises exactly `**22,785` parameters**:

- **Lightweight Byte Encoder:** Vocabulary size 256, model dimension $d_{\text{model}} = 32$, dual 1D convolutional layers with GELU activations and LayerNorm.
- **Thermodynamic Value Head:** Joint projection dimension $d_{\text{hidden}} = 64$, mapping joint state-action tokens to scalar reward $r_{\theta}(x, y) = -\log(1 + E(x, y))$.
- **Sub-Millisecond Inference:** Executes in 0.78 ms on CPU, enabling high-throughput candidate pre-filtering before invoking the sandbox.

6.2 DPO Formulation on Empirical Telemetry

Given prompt x , winning candidate y_w , and losing candidate y_l , the objective maximizes the log-likelihood margin:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x, y_w, y_l)} [\ln \sigma(\beta(r_{\theta}(x, y_w) - r_{\theta}(x, y_l)))] \quad (10)$$

All preference pairs are gathered from actual sandbox execution receipts (`DPORecord` schema). No synthetic approximations or fabricated latencies are permitted.

6.3 Empirical Telemetry Across Multidisciplinary Domains

Training over 25 epochs on multidisciplinary benchmarks (Rust numeric computing, pure mathematics, theoretical physics, and complex Python) yields:

- **DPO Loss:** Decreased by 20.10% ($0.6937 \rightarrow 0.5543$).
- **Reward Margin:** Increased from -0.0109 to $+3.0229$ (margin gain: $+3.0338$).
- **Computational Speedup:** Average speedup of $475.25\times$, reaching up to $4062.5\times$ on vectorizable numerical kernels.
- **Physical Energy Reduction:** Average energy reduction of 89.31%.

TABLE 3
Empirical Reinforcement Learning Optimization Across Domains

Case ID	Domain	Base (ms)	Opt (ms)	Speedup
PYTHON-01	complex_python	124.3	41.44	3.0×
RUST-26	rust_numeric	10.0k	63.11	158.4×
RUST-16	rust_numeric	10.0k	14.08	710.1×
MATH-28	pure_math	15.0	0.01	2182.3×
RUST-24	rust_numeric	10.0k	34.96	286.0×
MATH-24	pure_math	15.2	0.16	93.9×
RUST-18	rust_numeric	10.0k	7.91	1263.4×
PHYS-30	pure_physics	20.0	0.03	666.5×
PHYS-24	pure_physics	20.0	0.04	510.5×
PYTHON-26	complex_python	43.7	14.57	3.0×
PHYS-12	pure_physics	20.1	0.05	399.2×
PHYS-18	pure_physics	20.0	0.03	689.0×
PYTHON-30	complex_python	18.2	0.17	105.7×
PYTHON-28	complex_python	18.0	0.03	651.6×
MATH-25	pure_math	15.0	0.00	3956.7×
MATH-02	pure_math	870.1	310.76	2.8×

7 CONTEXT MANAGEMENT, EPISTEMIC REVIEW & REBUILD

7.1 Large-Window Context Management

ANSE avoids context exhaustion through four mechanisms:

- 1) **AST Skeletonization:** Reduces code token footprints by 84% by extracting interface definitions and invariant contracts.
- 2) **Ephemeral Scratchpad Isolation:** Offloads intermediate trace outputs and benchmark dumps out-of-context into disk scratchpads.
- 3) **Redis Hierarchical LTM:** Preserves sessions, conversation turns, and mathematical plans in Redis under `antigravity:conversation:*` and `antigravity:physics:*`.
- 4) **Epistemic Routing:** Isolates high-level planning to large reasoning models (Gemini 3.1 Pro) while execution runs on low-latency kernels (Gemini 3.8 Flash).

7.2 Low-Tier Directives (D1–D8) & Hardness Gates

To eliminate failure modes on resource-constrained reasoning models, ANSE implements Directives D1–D8:

- **D1 (Compressed Pain Prompts):** Strips verbose execution dumps to concise AST error spans (< 100 lines).
- **D2 (Capacity Gating):** Adaptively halts unproductive retry branches based on token consumption.
- **D3 (Fail-Fast Early Stopping):** Halts iterations immediately upon catastrophic syntax failure ($E \geq 10^6$) or diverging loss.
- **D4 (Skeleton Lessons):** Extracts interface-only learnings for long-term memory insertion.
- **D5–D8 (Tier Classification & Live Swapping):** Selects prompt strategies adaptively and executes autopoietic runtime swapping.

8 ANTI-HALLUCINATION NUMERIC EXECUTION HARNESS

The Anti-Hallucination Numeric Execution Harness (`paper_harness.py`) eliminates floating-point fabrication in academic papers:

Mandatory Sandbox Execution: All numbers in Tables 1, 2, and 3 are retrieved from executed processes and cryptographically hashed into numeric Receipt objects.

Section Partitioning: Generates bounded, isolated subsections preventing attention degradation.

Live Reference Grounding: Fetches external literature from arXiv over HTTPS, ensuring zero hallucinated citations.

9 CONCLUSION & OPEN GRAND CHALLENGES

ANSE establishes an empirical and mathematical foundation for autonomous artificial intelligence. By binding neural generation to the thermodynamic physics of computation, formal verification in Lean 4, and empirical reinforcement learning, ANSE eliminates phantom completions and numeric hallucinations.

We clearly delineate the boundary of what has been achieved:

- **Achieved:** Automated, closed-loop invariant verification and autotuning for complex mathematical specifications under hardware constraints.
- **Open Challenge:** Autonomous scientific discovery—the ability of an AI system to formulate novel conservation laws and hypothesize new physical equations without human specification.

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